MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER MG4DSCFNN200 – PRINCIPLES OF FOOD SCIENCE IN HOTEL OPERATIONS

PART A

1. Slows down microbial growth and extends shelf life of the foods
2. They prevent spoilage by inhibiting the growth of microorganisms
3. Bacteria, fungi(yeast and mould)
4. Reduce food wastage, makes food available during off-seasons
5. It is a sensory evaluation method in which trained panelists describe and measure sensory attributes of food quantitatively
6. Preference is the degree of liking or choice of one food product over another by consumers
7. Texture is the physical feel or structure of food perceived by touch, mouth feel, or chewing
8. The continuous phase is the medium in which dispersed particles are distributed
9. Encapsulation is the process of enclosing active ingredients within a protective coating
10. Foam is colloidal system in which gas bubbles are dispersed in a liquid or solid
11. Biopolymers of edible films are – polysaccharides, protein(gelatin), waxes
12. A coarse suspension is the heterogeneous mixture in which large solids particles are dispersed in a liquid and settle on standing
13. CAP-Controlled Atmospheric Packaging – fruits and vegetables, meat and meat products, sea foods and fishes, dairy products, grains nuts and seeds, processed and ready-to-eat foods.
14. Natural foods are food obtained directly from plants or animals with minimal processing and without artificial additives
15. A gel is a semisolid colloidal system in which liquid is trapped within a three dimensional solid network

PART B

16. Process of canning

- Selection of good quality raw materials
- Washing and preparation
- Blanching
- Filling in to cans
- Addition of syrup or brine
- Exhausting (removal of air)

- Sealing
- Sterilization by heat
- Cooling
- Storage

Causes of spoilage in canned foods

- Incomplete sterilization
- Leakage in cans
- Chemical reaction with metals
- Survival of thermophilic bacteria
- Swelling due to gas production

17. Food irradiation is the treatment of food using ionizing radiation such as gamma rays, X-rays, or electron beams to destroy microorganisms and insects.
    Why called cold sterilization:
    - It destroys microbes without raising food temperature significantly
    - Nutritional quality is largely preserved
    - Suitable for heat-sensitive foods

    Uses: Extends shelf life, Prevents sprouting, Controls insects

18. Paired comparison test is a sensory evaluation method in which two samples are presented together and the judge compares them for a specific characteristic.
    Procedure:

    - Two coded samples are given
    - Judge compares for sweetness, saltiness, texture, etc.
    - Judge selects sample with stronger characteristic

19. Sensory evaluation is the scientific method of judging food by sense organs such as taste, smell, sight, touch, and hearing.
    Applications:Product development, Quality control, Consumer acceptance studies, Shelf-life evaluation, Detecting defects in food
20. Emulsion is a mixture of two immiscible liquids where one liquid is dispersed in another.
    Process of emulsion formation:

    - Two liquids (oil and water) are mixed
    - Mechanical agitation breaks one liquid into droplets
    - Emulsifier stabilizes droplets

    Types:

    - Oil in water (milk)
    - Water in oil (butter)

21. Modified Atmosphere Packaging (MAP) is a packaging technique in which the normal air inside the package is replaced with a controlled gas mixture.
Common gases used: Oxygen, Carbon dioxide, Nitrogen

Advantages:

- Extends shelf life
- Reduces microbial growth
- Maintains freshness

PART C

22. Food preservation is the process of preventing food spoilage and extending shelf life by controlling microorganisms, enzymes, and chemical changes.
Objectives:

- Prevent microbial spoilage
- Maintain nutritional quality
- Extend shelf life
- Ensure safety

Importance:

- Reduces food wastage
- Makes seasonal foods available year-round
- Facilitates transport and storage
- Improves food security

Need in modern society:

- Urbanization increases demand for stored foods
- Long-distance food transport
- Emergency food supply
- Convenience foods for busy lifestyles

Methods

- High temperature preservation
- Low temperature preservation
- Dehydration
- Irradiation
- Preservatives

23. Aseptic packaging is filling commercially sterile food into sterile containers under sterile conditions.
    Steps:

    1. Sterilization of food (UHT treatment)
    2. Sterilization of packaging material
    3. Sterile filling
    4. Hermetic sealing
    5. Cooling and storage

    Advantages:

    - Long shelf life
    - No refrigeration needed before opening
    - Better nutrient retention

    Examples:

    - Milk packs
    - Fruit juices

24. Health foods are foods that provide additional nutritional benefits and help maintain good health.
    Examples:

    - Whole grains
    - Fruits and vegetables
    - Sprouts
    - Low-fat dairy products
    - Nuts

    Benefits:

    - Improve immunity
    - Prevent lifestyle diseases
    - Provide essential nutrients
    - Aid digestion
    - Maintain body weight